Adarsh Bajpai

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: github.com/nameadarsh | : nameadarsh.github.io/Portfolio

Summary

AI Engineer building real-world machine learning systems and pipelines from development to deployment, with a focus on performance, reliability, and practical impact.

Experience

AI Engineer — **Jan 2026 – Present**
Global Infoventures Pvt. Ltd. — *Delhi, India*

- Developed a RAG pipeline for document querying and question answering with support for Hindi text; used chunking strategies to preserve context and improve response quality.
- Contributed to the development of a scalable Learning Management System (LMS) as part of a joint project with TCS, to serve a student base of **2 lakh users** across multiple universities.

Projects

Leaf Disease Detection System **[GitHub]** | *Python, U2-Net, ResNet, Flask, React, OpenCV, NLP*

- Built a hybrid deep learning model **(U2-Net + ResNet)** for crop disease detection and severity estimation, achieving **96% classification accuracy** and reducing misclassification rate by **∼15%**.
- Extended into a full-stack web app **(React + Flask)** with image upload, bilingual results, severity scoring, and an **LLM chatbot** for real-time farmer queries.

Real-Time Multi-Object Tracking Pipeline **[GitHub]** | *RT-DETR, ByteTrack, OpenCV, Pandas, PyYAML*

- Built an end-to-end tracking pipeline using **RT-DETR-R50 and ByteTrack**, designed for **Indian road environments**; detects and **tracks vehicles, pedestrians, and animals** with stable persistent IDs across video streams.
- Processes videos in batch mode, **producing annotated outputs**, side-by-side comparisons, and per-detection **CSV logs with confidence scores**. Pipeline behavior is configurable via YAML (tracking thresholds, parameters, I/O settings).

Sanity - Real vs. Fake News Detection **[GitHub]** | *DistilBERT, Flask, React, NLP, Groq LLM*

- Developed an AI-powered news classification system using a **fine-tuned DistilBERT model** supporting **text, PDF, and URL inputs.** When the confidence threshold is below 70%, it automatically triggers an **LLM fallback using Groq API** with NLP-processed context for inference.
- Designed an NLP preprocessing pipeline (tokenization, lemmatization, stopword removal) and an **interactive Q&A system for context-aware article queries**. Deployed end-to-end using Flask and React.

Technical Skills

Programming: Python, JavaScript

Data Science & ML Libraries: PyTorch, NumPy, Pandas, Matplotlib, OpenCV

AI & Machine Learning: Machine Learning, CNNs, ResNet, U2-Net, Vision Transformers, RT-DETR, ByteTrack, Natural Language Processing, RAG Pipelines, LLM Integration, Data Preprocessing, Feature Engineering

Web Frameworks & Development: Flask, FastAPI, React, Node.js, MongoDB

DevOps & Tools: Git, GitHub, Docker, Jenkins, Postman, Jira

Design & Visualization: Figma, Wireframing, UI Design, Prototyping, Adobe Suite

Education

Bachelor of Technology in Computer Science — **Aug. 2022 – May 2026**
NIIT University | *AI and ML Specialisation* — *Neemrana, Rajasthan*

- Coursework: Machine Learning, Generative AI, AI Agents and Systems, Software Engineering, DBMS

Certifications

Oracle Cloud Infrastructure 2025 Generative AI Professional **[Certificate]**
Oracle | *Generative AI, model deployment on OCI, prompt engineering, enterprise AI integration*

React Web Developer Certificate **[Certificate]**
Infosys | *Full-stack React development, backend APIs, routing, state management, production-ready web apps*